

Single Source Justification (Simplified Acquisition)

Section A: General Contract Information

Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan](#)?

☐ Yes

☒ No

Contracting Activity: AFRL/PZLET

Project/Program Name: Aerospace Vehicles Computational Facilities Support

Purchase Request / Local ID Number:

Estimated Contract Cost (including Options):

Section B: Description of the Supplies/Services Required

The Design and Analysis Branch of the Aerospace Systems Directorate of the Air Force Research Laboratory (AFRL / RQ) has a requirement to access and use HPC resources among members of the community (RQV). Local HPC Resources were acquired and on-boarded around 2021. These resources are system administered along with other HPC resources within AFRL/IZ.

The RQHAP cluster is required by Security services to move to a newer version of the core OS 9 compatible RHEL 8.10 kernels). Before this can be done components that make up the storage servers require maintenance support to their latest versions so they are compatible and able to receive the RHEL kernel updates.

Section C: Justification for Soliciting from a Single Source or Brand Name

In accordance with RFO 12.102(a), the HAP system is a highly specialized High-Performance Computing (HPC) cluster utilizing a storage architecture designed and manufactured by DataDirect Networks (DDN). The required maintenance involves a critical update to the storage server firmware and software to ensure essential compatibility between Lustre clients and RHEL 8.10 kernels. DDN Federal, LLC retains proprietary rights to the source code, internal firmware, and diagnostic tools required to perform these updates. No other vendor has the legal access or technical capability to modify the DDN proprietary storage environment without risking system instability or total loss of data integrity. Furthermore, the HAP system is currently under a 3-Year BSPO Support agreement (SO# 125514) ending 2/25/2026. Per the terms of the original equipment manufacturer (OEM), the introduction of third-party, non-certified technicians to perform core system updates would void existing support agreements and hardware warranties.

DDN Federal, LLC is the only source with the certified Professional Services personnel authorized to perform these specific version updates while maintaining the system's operational certification within the AFRL/IZ and RQV environments. To maintain the mission-essential functions of AFRL/RQV, the cluster must transition to RHEL 8.10. Failure to utilize DDN-specific services would result in a compatibility mismatch between the Lustre file system and the operating system kernels, effectively rendering the HAP cluster inaccessible for research activities. Because the storage architecture is a closed-loop proprietary system, there are no equivalent services available that can guarantee the necessary interoperability. This acquisition will be funded using FY26 3600 funds.

As mentioned previously DDN is the proprietary maker of the storage firmware and software. Any bid that would have proprietary knowledge of what the storage firmware and software need would require any entity adding in the

Single Source Justification (Simplified Acquisition)

maker of those tools/system, DDN, to avoid catastrophic loss of functionality. This would have the Government entering an effort that would consist of the known advertised price provided in the quote by DDN, plus any fees and requirements of a potential bid awardee that would sub the efforts through DDN. Any scenario where we do not have DDN directly or indirectly involved risks leaving HAP in a state where it is non-functional stalling the research efforts of our mission and incurring unrecoverable costs.

Loss of compute calculations and examinations translates to loss of either some iteration or part of a final product. The final product in this case are improvements and enhancements to Air Vehicles and methods to allow our war-fighters to fly further, hit harder, etc. Loss of HAP processing affects our mission to help equip our war-fighters with the edge needed to win the future.

It is for these reasons above that AFRL/RQVC believes that bidding for an entity outside of DDN would incur significant cost and resource drain on the Government. Time and money lost in this effort would be through a competition and alternative bid effort.

Section D: Efforts to Obtain Competition

Market research was conducted to identify potential sources capable of performing the required maintenance and firmware updates for the HAP system's DataDirect Networks (DDN) storage architecture. This research included a review of the original equipment manufacturer (OEM) technical specifications and an inquiry into the existing support agreement. It was determined that because the HAP cluster utilizes proprietary DDN firmware and a specialized Lustre configuration, only DDN Federal, LLC possesses the authorized access to the source code and diagnostic tools necessary to ensure compatibility with RHEL 8.10. While other third-party IT service providers exist in the general marketplace, none are authorized by the OEM to perform these specific system-level modifications. Furthermore, utilizing a non-OEM source would result in the immediate voiding of the existing 3-Year BSPO support and hardware warranties, creating an unacceptable financial and operational risk to

Single Source Justification (Simplified Acquisition)

the government. Consequently, DDN Federal, LLC was identified as the only source reasonably available to satisfy this requirement without causing a lapse in system integrity or support coverage.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

AFRL/RQ will continue to monitor the high-performance computing (HPC) and data storage markets by reviewing technical catalogs, conducting internet research, and engaging with industry experts at technical trade shows and conferences. For the current HAP system, the government is restricted by the proprietary nature of the existing DataDirect Networks (DDN) architecture and the associated warranty requirements. However, as this system approaches its end-of-life or as future expansion requirements are identified, AFRL/RQ will perform a thorough analysis of alternative open-standard storage solutions. Every effort will be made to ensure that future hardware and software acquisitions are structured to allow for full and open competition among all qualified vendors, provided those solutions can meet the rigorous technical and interoperability standards required by the AFRL/IZ and RQV mission sets.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.